Trichotomy for the membership ordering of ordinals: any two ordinals satisfy one of alpha in beta, alpha = beta, beta in alpha

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ordinals/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: both induction layers, the exclusion of the two membership alternatives, and each inclusion are written out in the span; the only implicit input, transitivity of an ordinal, is named in-line by the author.

Editorial note. The label ordtrichotomy names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Transfinite induction schema for ordinals. Same unit, \ollabel{ordinductionschema} (source line 446). Used twice, as the double induction the proof opens with (line 484).
- Elements of ordinals are ordinals. Same unit, \ollabel{ordmemberord} (source line 416). At line 497: gamma in alpha makes gamma an ordinal, so the first induction hypothesis applies to it.
- Transitivity of ordinals. The unit's Definition/earlier facts on ordinals; invoked in-line at 499-501 with the parenthetical 'invoking the fact that alpha is transitive if necessary'. Turns gamma = beta or beta in gamma into beta in alpha, contradicting the standing assumption.

Source reference. Original reference ordinductionschema: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 446–446. Prerequisite pointer invoked at proof line(s) 484; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ordmemberord: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 416–416. Prerequisite pointer invoked at proof line(s) 497; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1 (Trichotomy)**.** *$\alpha \in \beta \vee \alpha = \beta \vee \beta \in \alpha$, for any ordinals α and β.*

Proof. The proof is by double induction, i.e., using ordinductionschema twice. Say that x is *comparable* with y iff $x \in y \vee x = y \vee y \in x$.

For induction, suppose that every ordinal in α is comparable with *every* ordinal. For further induction, suppose that α is comparable with every ordinal in β. We will show that α is comparable with β. By induction on β, it will follow that α is comparable with every ordinal; and so by induction on α, *every* ordinal is comparable with *every* ordinal, as required. It suffices to assume that $\alpha \notin \beta$ and $\beta \notin \alpha$, and show that $\alpha = \beta$.

To show that $\alpha \subseteq \beta$, fix $\gamma \in \alpha$; this is an ordinal by ordmemberord. So by the first induction hypothesis, γ is comparable with β. But if either $\gamma = \beta$ or $\beta \in \gamma$ then $\beta \in \alpha$ (invoking the fact that α is transitive if necessary), contrary to our assumption; so $\gamma \in \beta$. Generalising, $\alpha \subseteq \beta$.

Exactly similar reasoning, using the second induction hypothesis, shows that $\beta \subseteq \alpha$. So $\alpha = \beta$. □

As such, we will sometimes write $\alpha < \beta$